Lecture 1

Statements, sets and proofs

Uses.

School algebra only (syllabus section 2).

Objectives. *A mathematical claim specifies the objects it concerns and the property it asserts. We learn to read those specifications, negate them and prove them. The same rules will govern the definitions of a limit and continuity.*

By the end of this lecture you will be able to:

- Negate statements with implications and quantifiers over specified sets.
- Distinguish an implication from its converse and contrapositive.
- Prove a claim directly, by contraposition or by contradiction.
- Refute a universal claim by a counterexample in its stated domain.

1.1 Statements and their domains

A *statement* is a claim that is either true or false. The sentence "the integer 12 is divisible by 3" is a statement. The expression $x^2 > 1$ does not yet specify a statement. Its truth depends on the number assigned to x. For $x = 2$ it is true. For $x = 0$ it is false.

The *domain* specifies which objects a variable may denote. Throughout these notes, the natural numbers and integers are denoted by

$$\mathbb{N} = \{1, 2, 3, \ldots\}, \qquad \mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}.$$

The symbol $\mathbb{Q}$ denotes the rational numbers. A rational number has the form p/q, where $p \in \mathbb{Z}$ and $q \in \mathbb{N}$. The symbol $\mathbb{R}$ denotes the real numbers. We use their school arithmetic. Lecture 2 states the additional property of $\mathbb{R}$ that makes the existence of limits possible.

A *set* is a collection of objects, called its *elements*. The notation $x \in A$ means that x belongs to the set A. The notation $x \notin A$ means that it does not. The empty set, denoted by $\varnothing$, has no elements. For instance, the set

$$A = \{x \in \mathbb{R} : x^2 = 1\}$$

has the elements -1 and 1. The colon means "such that". The displayed condition determines which real numbers belong to A.

The same formula can make different claims on different domains. Every $n \in \mathbb{N}$ satisfies $n^2 \geq n$. The assertion that every $x \in \mathbb{R}$ satisfies $x^2 \geq x$ is false, since $x = 1/2$ violates it. A proof must use only the assumptions stated for its domain. A counterexample must belong to that domain.

We write $A \subseteq B$ if every element of A is an element of B. This is called *inclusion*. Equality of sets means that they have the same elements. To prove $A = B$, it is enough to prove $A \subseteq B$ and $B \subseteq A$. This method requires two implications. Checking one of them proves only an inclusion.

Intervals specify common real domains. For $a < b$, we use the notation

$$(a, b) = \{x \in \mathbb{R} : a < x < b\}, \qquad [a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}.$$

A round endpoint is excluded. A square endpoint is included. The intervals $(a, b]$ and $[a, b)$ include just one endpoint. An interval such as (a, ∞) means $\{x \in \mathbb{R} : x > a\}$, and $[a, \infty)$ includes a. Infinity is not a real endpoint or an element that the interval contains.

1.2 Combining statements

Let P and Q be statements. Their conjunction, written $P \wedge Q$, means that P and Q are both true. Their disjunction, written $P \vee Q$, means that at least one is true. Mathematical "or" includes the case where both are true. The negation $\neg P$ is true exactly when P is false.

An inequality illustrates the difference between a claim and its negation. For $x \in \mathbb{R}$, the negation of $x < 2$ is $x \geq 2$. It includes equality. The assertion $x > 2$ excludes equality, so it is not the negation of $x < 2$. This endpoint distinction will matter when we negate a strict error bound.

An *implication*, written $P \Rightarrow Q$, asserts that whenever P is true, Q is true. The statement P is its *hypothesis*. The statement Q is its *conclusion*. A failure of the implication means that its hypothesis holds but its conclusion fails. Its truth values are therefore given by

P	Q	$P \Rightarrow Q$
true	true	true
true	false	false
false	true	true
false	false	true

For example, for every $x \in \mathbb{R}$ the following implication holds:

$$x > 2 \implies x^2 > 4.$$

A number $x \leq 2$ cannot refute it. At such a number its hypothesis does not hold. To refute the implication, we would need a real number that is larger than 2 and whose square is at most 4. No such number exists: if $x > 2$, then both $x - 2$ and $x + 2$ are positive, so $x^2 - 4 = (x - 2)(x + 2) > 0$.

The *converse* of $P \Rightarrow Q$ is $Q \Rightarrow P$. The converse makes a different claim. The number $x = -3$ shows that $x^2 > 4$ does not imply $x > 2$. Knowing that a square is large does not determine the sign of its base.

The *contrapositive* of $P \Rightarrow Q$ is $\neg Q \Rightarrow \neg P$. An implication and its contrapositive have the same truth value. Both fail exactly when P is true and Q is false. For the preceding example, the contrapositive is

$$x^2 \leq 4 \implies x \leq 2.$$

It retains the original direction of reasoning in its logically equivalent form. It is not the converse.

An *equivalence*, written $P \Leftrightarrow Q$, asserts both $P \Rightarrow Q$ and $Q \Rightarrow P$. We also say "P if and only if Q". Proving an equivalence requires both directions unless a chain of already justified equivalences establishes them together.

Proposition 1.1 (Negation rules). *For statements P and Q, negation obeys the rules*

$$\neg(P \wedge Q) \;\Leftrightarrow\; (\neg P) \vee (\neg Q), \qquad \neg(P \vee Q) \;\Leftrightarrow\; (\neg P) \wedge (\neg Q).$$

The negation of $P \Rightarrow Q$ is $P \wedge \neg Q$.

Proof. The conjunction $P \wedge Q$ fails exactly when at least one of P and Q is false. The disjunction $P \vee Q$ fails exactly when both are false. The implication table has one false case: P is true and Q is false. These are the stated negations. ∎

For a fixed $x \in \mathbb{R}$, negating the condition $0 < x < 1$ therefore gives $x \leq 0$ or $x \geq 1$. Negating the claim that an implication holds means retaining its hypothesis and negating its conclusion.

The same connectives define set operations. The union $A \cup B$ contains the objects belonging to A or B. The intersection $A \cap B$ contains the objects belonging to both. The difference $A \setminus B$ contains the elements of A that do not belong to B. These definitions specify membership, so an identity between such sets can be proved by following an arbitrary element through its membership conditions.

1.3 Quantifiers and their negations

A *universal quantifier* asserts a property of every object in a domain. We write "for every $x \in A$" as $\forall x \in A$. An *existential quantifier* asserts that at least one object in the domain has the property. We write "there exists $x \in A$" as $\exists x \in A$. Existence does not assert uniqueness.

A proof of a universal statement must cover an arbitrary member of its domain. A proof of an existential statement can give one object and verify its property. We call such an object a *witness*. These are different proof obligations.

Example 1.2 (Order of quantifiers). We prove the statement

$$\forall x \in \mathbb{R} \quad \exists y \in \mathbb{R} : \quad y > x.$$

Given $x \in \mathbb{R}$, choose $y = x + 1$. This is a real number and satisfies $y > x$. The choice may depend on x, because x was given first.

Reversing the quantifiers gives the statement

$$\exists y \in \mathbb{R} \quad \forall x \in \mathbb{R} : \quad y > x.$$

This statement is false. For any proposed $y \in \mathbb{R}$, the real number $x = y + 1$ satisfies $x > y$. Thus no single y works for every x.

The order of choices is part of a quantified claim. In the first statement, the witness y is chosen after x is specified. In the second statement, it must be chosen before x is specified. The symbols alone do not record a particular formula for the witness. A proof supplies that formula or another justified construction.

Negating a universal statement requires a counterexample. Negating an existential statement requires excluding every possible witness. The quantifier rules are

$$\neg(\forall x \in A :\; P(x)) \;\Leftrightarrow\; \exists x \in A :\; \neg P(x), \tag{1.1}$$

$$\neg(\exists x \in A:\ P(x)) \iff \forall x \in A:\ \neg P(x).$$

For a claim with several quantifiers, apply these rules in their original order. Each quantifier changes kind. The final property is negated too.

For instance, negating the first displayed statement of the example gives

$$\exists x \in \mathbb{R} \quad \forall y \in \mathbb{R}: \quad y \le x.$$

This would assert a real number at least as large as every real number. The statement is false, in agreement with the proof of the original claim. The relation changes from $y > x$ to $y \le x$, rather than to $y < x$.

The earlier implication about squares was asserted for every $x \in \mathbb{R}$. Its negation must therefore include an existential quantifier:

$$\exists x \in \mathbb{R}: \quad x > 2 \text{ and } x^2 \le 4.$$

This asks for one real counterexample satisfying the hypothesis and failing the conclusion. Negating the implication alone describes the property that such a counterexample would have. Negating the universal claim also asserts that a counterexample exists.

A domain restriction must survive negation. The negation of "every $x \in (0,1)$ satisfies $x^2 < x$" asks for an $x \in (0,1)$ with $x^2 \ge x$. The number $x = 2$ cannot be that witness because it lies outside the domain. When the domain is empty, a universal statement has no counterexample and is true. An existential statement over the empty set is false because it has no possible witness.

1.4 Direct proof and contraposition

A *proof* is an argument that derives a claim from its stated assumptions by valid steps. In a direct proof of $P \Rightarrow Q$, we assume P and derive Q. We do not assume the conclusion we are trying to establish.

An integer n is *even* if $n = 2k$ for some $k \in \mathbb{Z}$. It is *odd* if $n = 2k+1$ for some $k \in \mathbb{Z}$. Every integer is exactly one of even or odd, by division with remainder by 2. We use this school arithmetic fact.

Proposition 1.3 (Parity of a square). *For $n \in \mathbb{Z}$, the integer n^2 is even if and only if n is even.*

Proof. First suppose n is even. There exists $k \in \mathbb{Z}$ such that $n = 2k$. Then $n^2 = 4k^2 = 2(2k^2)$. Since $2k^2 \in \mathbb{Z}$, this is the definition of an even integer.

For the converse, we prove the contrapositive. Suppose n is odd, so $n = 2k+1$ for some $k \in \mathbb{Z}$. Expansion gives

$$n^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

The number $2k^2 + 2k$ is an integer. Thus n^2 is odd. We have proved that an odd integer cannot have an even square. Hence an integer with an even square is even. ∎

The two directions use different arguments. In the direct direction, substituting the even form of n produces the even form of n^2. In the other direction, the odd form of n produces an odd square. The latter calculation proves a contrapositive and therefore proves the required implication.

A true converse cannot be inferred from a true implication without an argument. In this proposition it has an argument. The earlier example $x > 2 \Rightarrow x^2 > 4$ had a false converse. The difference lies in the properties being proved, not in a rule that reverses every implication.

A *counterexample* refutes a universal claim by satisfying its hypotheses and violating its conclusion. One example suffices to refute such a claim. Any finite list of successful examples proves only those instances. It does not cover all members of an infinite domain.

A proof can use an arbitrary element without assigning it a numerical value. For example, when n is an arbitrary even integer, the expression $n = 2k$ covers every even integer because it states their defining property. The calculation above is therefore universal. Checking only $n = 2$, $n = 4$ and $n = 6$ would not have that effect.

1.5 Proof by contradiction

In a *proof by contradiction*, we assume the hypotheses and the negation of the desired conclusion. We derive an impossibility from these assumptions. Since the hypotheses are retained, the negated conclusion must fail.

We use positive square roots with their school meaning. The number $\sqrt{2}$ is the positive real number whose square is 2.

Theorem 1.4 (Irrationality of $\sqrt{2}$)**.** *The number $\sqrt{2}$ is not rational.*

Proof. We argue by contradiction. Suppose $\sqrt{2} = p/q$, where $p \in \mathbb{Z}$ and $q \in \mathbb{N}$. By cancelling common factors, choose this representation with p and q having no common integer divisor greater than 1. Squaring the equality and multiplying by q^2 gives

$$p^2 = 2q^2.$$

Hence p^2 is even. By Proposition 1.3, p is even. Write $p = 2k$ with $k \in \mathbb{Z}$. Substitution gives

$$4k^2 = 2q^2, \qquad q^2 = 2k^2.$$

Thus q^2 is even. The same proposition implies that q is even. The integers p and q therefore share the divisor 2. This contradicts the chosen representation. Hence $\sqrt{2} \notin \mathbb{Q}$. ∎

The contradiction is the common divisor of a fraction assumed to have none. The equality $p^2 = 2q^2$ is an intermediate consequence, rather than a contradiction by itself. Keeping track of the assumed properties of p and q identifies the final step.

A contradiction proof and a contrapositive proof are related but have different starting assumptions. To prove $P \Rightarrow Q$ by contraposition, we assume $\neg Q$ and derive $\neg P$. By contradiction, we assume both P and $\neg Q$ and derive an impossibility. Both establish the original implication.

The irrationality proof used the least-terms representation of a rational number and the parity proposition. It did not assume any decimal expansion of $\sqrt{2}$. An approximation such as 1.414 determines neither the existence of a rational representation nor its impossibility. The proof addresses every proposed rational representation at once.

In later lectures, an existence claim and a uniqueness claim will often appear in the same theorem. They require separate arguments. Existence asks for at least one object satisfying the stated conditions. Uniqueness asks that any two objects satisfying them be equal. A contradiction proof may establish uniqueness by assuming two distinct candidates. It does not supply the existence argument.

Summary 1.

- A universal statement concerns every member of its specified domain. An existential statement requires at least one witness in that domain.
- Proposition 1.1 and Eq. (1.1) give negations of connectives, implications and quantified statements. Negating an implication retains its hypothesis and negates its conclusion.
- An implication is equivalent to its contrapositive. Its converse needs a separate proof. An equivalence asserts both directions.
- Proposition 1.3 and Theorem 1.4 illustrate direct proof, contraposition and contradiction. The number $\sqrt{2}$ is irrational.

A limit definition will specify the order in which an accuracy and an index are chosen. The quantifier rules here allow us to state exactly when such a definition fails.

References

R. G. Bartle and D. R. Sherbert, *Introduction to Real Analysis*, 4th ed., Wiley, 2011, §1.1 and Appendix A. P. J. Cameron, *Introduction to Algebra*, 2nd ed., Oxford University Press, 2008, §§1.1–1.2 and 1.26–1.27.